

Refactoring the GRC pipeline

What changed in RP7, and why

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Pipeline was unwieldy with 22 .do files + lots of duplication

- 22 numbered .do files; experience and birth covariates spread across six of them
- Parallel Dropbox folder for real-values work, required manual “syncing”
- `.ster` filenames (Stata results files) collided across .do files; some exceeded Stata's name limit

All of this makes the pipeline hard to maintain.

Fewer .do files and programs to keep track of

Before (RP6, 22 files)

- 5_GrRC.do
- 6_GrRC_NonAg.do
- 8_GrRC_hukou.do
- 10_GrRC_experience.do
- 11_GrRC_max_experience.do
- 12_GrRC_experience_share.do
- 13_GrRC_max_experience_share.do
- 14_GrRC_NonAg_experience.do
- 15_GrRC_birth.do
- 16_heterogeneity_tables.do
- ...

After (RP7, 16 files)

- 1_processData.do
- 2_summaryStats.do
- 3_OLS_uGRC.do
- 4_GrRC.do
- 5_GrRC_NonAg.do
- 6_OLS_uGRC_hukou.do
- 7_GrRC_hukou.do
- 8_learning.do
- 9_GRC_extras.do
- 10_make_tables.do,
- 11_make_figures.do

Adding a new covariate (or set of covariates)

Before

- Each new covariate got its own .do file: `10_GrRC_experience.do`, `11_GrRC_max_experience.do`, ..., `15_GrRC_birth.do`

Now

- All extra-covariate regressions live in `9_GRC_extras.do`
- One line per (country, spec, covariate)
- Tables: `extras_tex_table` reads `.ster` files and writes the corresponding `.tex` tables

To add a new covariate: add a line in `9_GRC_extras.do`.

One table-builder for all GRC tables

We had multiple table-building programs that did 95% the same thing:

- `grc_tex_table_trend`
- `grc_tex_table_trend_exp`
- `grc_tex_table_trend_birth`
- `grc_tex_table_trend_hukou`

Now: one `grc_tex_table_trend`, four call signatures.

To change column widths, indicator-row order, postfoot string, or decimal precision, edit `grc_tex_table_trend` in `0_programs.do`.

One switch handles nominal and real values

Before

- Two parallel Dropbox folders
- Every edit propagated to both, by hand

Now

- One global, `values(nominal|real)`, in `0_path_config.do`
- **nominal**: filenames as before
- **real**: appends `_r` suffix; both save to the same `output/` folder

To switch modes: edit global `values` in `0_path_config.do`.

How to read a .ster filename

Pattern: `grc_<country>_<spec3>_<covs2>[_<sfx1>].ster`

country	IDN, CHN, TZA, or CHN_hukou_<subgroup>
spec3	cuu, cub, iuu, cnu
covs2	c0, ct, c1, c2, ca
sfx1	n (never), a (always), d (delta), g (average)

Every .ster name is built from these tokens. Look up any fit by combining them.

Full convention in `RP7/scripts/STER_NAMING.md`.

How to edit GRC tables in the new pipeline

Caption, label, or table notes (now in the paper, not the .do file)

- Edit the `\GRCtable{}` and `\GRChukoutable{}` macros in `preamble.tex`

Which coefficients appear, label text, decimal formatting

- Per-cell options: `10_make_tables.do`
- Underlying template: `grc_tex_table_trend` in `0_programs.do`

Cosmetic refresh

- Average Δ row reads $\bar{\Delta}$
- Empty `esttab` rows now stripped

What this buys us

- Regression code lives in one place: `4_GrRC.do` for the main regressions, `9_GRC_extras.do` for extras
- Table caption, label, and notes live in one place: the paper preamble
- `.ster` filenames you can read, type, and look up

What each .do file does

Setup (run automatically by 0_master.do)

0_master.do

0_path_config.do

0_programs.do

0_setup.do

0_CHN_hukou_restrictions.do

runs all the .do files in sequence

paths, plus the **values(nominal|real)** switch

shared programs (**run_grc**, table builders, helpers)

Stata environment, package installs

CHN datasets with hukou subgroup definitions

Pipeline sequence

1_processData.do

2_summaryStats.do

3_OLS_uGRC.do

4_GrRC.do

5_GrRC_NonAg.do

6_OLS_uGRC_hukou.do

7_GrRC_hukou.do

8_learning.do

9_GRC_extras.do

10_make_tables.do

11_make_figures.do

reads raw **.dta**, builds processed datasets

summary statistics tables

OLS and FE regressions, all countries

restricted GRC regressions, all countries

GRC regressions with non-ag as choice (IDN)

OLS and FE regressions, hukou subgroups (CHN)

GRC regressions, hukou subgroups (CHN)

returns to learning, rural and urban

GRC with extra covariates (experience, birth)

builds **.tex** tables from **.ster** files

builds figures